5. Si3218x Hardware Design Guide

This section contains the important guidelines for a successful hardware implementation including reference schematics, bills of materials, and layout guidance. These guidelines should be followed to ensure the Si3218x device provides the best performance possible.

Guidelines are divided in the following sections:

- Si32184/5 Top-Level Design
- Si32182/3 Top-Level Design
- Low-Cost Capacitive Boost Design
- Surge and AC Mains Protection Options

5.1 Si32184/5 Top-Level Design

This design features the PCM/SPI enabled Si32184/5 device in a 6x6 mm 40-pin QFN package, which is shown in the figure below. Design considerations for this option include:

- Supply decoupling networks
- AC and DC sense networks
- DC-DC charge pump
- Bias network components
- Host Interface
- Optional power cross diode

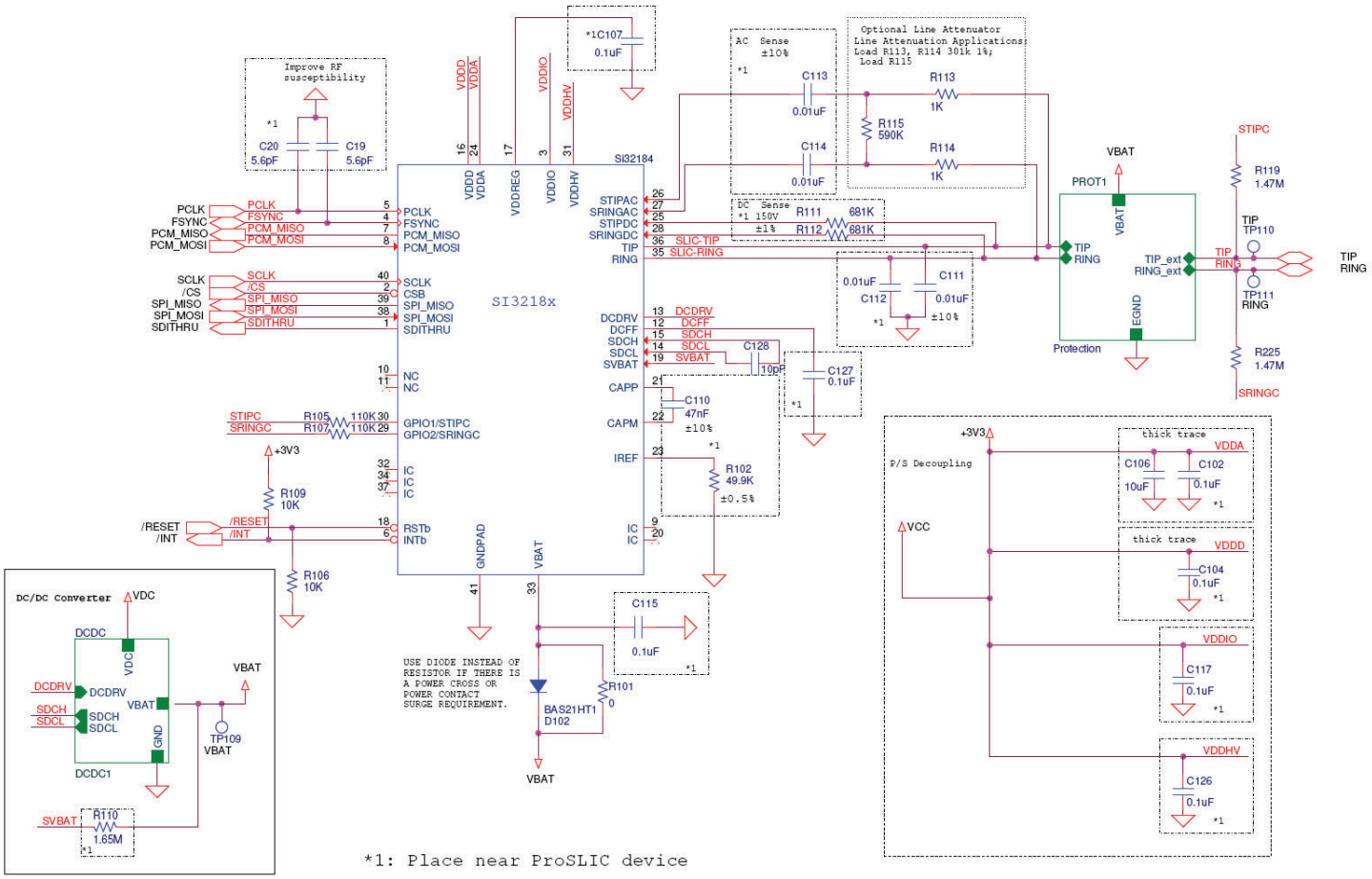


Figure 5.1. Si32184/5 Top-Level Schematic

5.1.1 Si32184/5 Top-Level Bill of Materials

The table below is a comprehensive bill of materials (BOM) for the top-level Si32184/5 design.

Table 5.1. Si32184/5 Top-Level Bill of Materials

Quantity	Component	Description	Value	Rating	Accuracy	Type	Package
5	C102,C104,C107,C117,C126	Decoupling capacitor	0.1 µF	10 V	±10%	X7R	0402
1	C106	Decoupling capacitor	10 µF	6.3 V	±20%	X5R	0603
1	C110	Speedup capacitor	47 nF	16 V	±10%	X7R	0402
2	C111,C112	Linefeed capacitor	0.01 µF	200 V	±10%	X7R	0805
2	C113,C114	AC Sense Capacitor	0.01 µF	200 V	±10%	X7R	0805
1	C115	VBAT decoupling capacitor	0.1 µF	250 V	±10%	X7T	0805
1	C127	DCDC FET drive charge pump capacitor	0.1 µF	10 V	±10%	X7R	0402
1	R101	Power cross diode bypass resistor	0	2 A		ThickFilm	0805
1	R102	Reference resistor	49.9 K	1/16 W	±0.5%	ThickFilm	0402
2	R105,R107	Coarse sense resistor	110 K	1/16 W	±1%	ThickFilm	0402
1	R109	Pull-up resistor	10 K	1/16 W	±5%	ThickFilm	0402
1	R110	VBAT sense resistor	1.65 M	1/10 W	±1%	ThickFilm	0805
2	R111,R112	DC sense resistor	681 K	1/10 W	±1%	ThickFilm	0805
2	R113,R114	AC sense resistor	1 K	1/10 W	±1%	ThickFilm	0603
2	R119,R225	Coarse sense resistor	1.47 M	1/8 W	±1%	ThickFilm	1206
1	U1	SLIC IC	Si32184	100 V		MCM	QFN40 6x6mm
2	C19,C201	Host digital I/O attenuation	5.6 pF	50 V	±0.25pF	NP0 HIGH Q	0402
1	D1022	Power cross diode	BAS21HT1	250 V/600 mA		Single	SOD-323
1	R1063	Pull-down resistor	10 K	1/16 W	±5%	ThickFilm	0402
1	R1154	AC attenuator resistor	590 K	1/10 W	±1%	ThickFilm	0805
1	C1285	Current sense attenuator capacitor	10 pF	50 V	±5%	C0G	0402

Notes:
1. Optional capacitors for noise control of host interface.
2. Optional diode if design must support power cross.
3. Optional pull-down resistor for /RESET. The /RESET pin must be held low until PCM clock is established and stabilized.
4. Optional resistor for AC attenuation.
5. Optional current sense attenuator capacitor; attenuates high-frequency noise in current sense lines.

5.1.2 Si32184/5 Top-Level Layout Guidelines

The following placement and/or routing guidelines should be followed when implementing the Si32184/5.

1. Place decoupling capacitors as close to the device as possible. Bulk decoupling capacitors may be placed farther away if necessary.

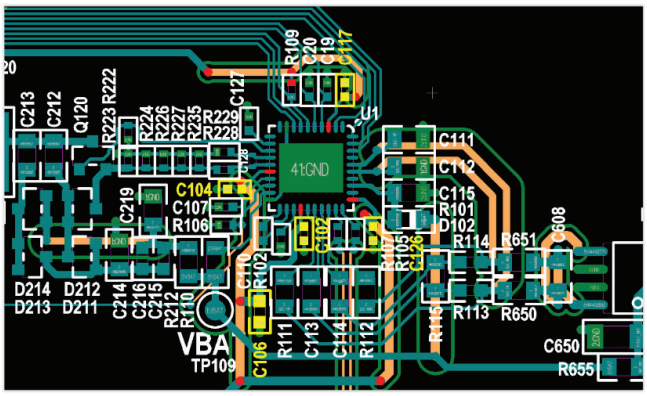


Figure 5.2. Decoupling Capacitor Placement

2. Place DCDC FET drive charge pump capacitor C127, speedup capacitor C110, VBAT decoupling capacitor C115, and Current sense attenuator capacitor C128 as close to device as possible.

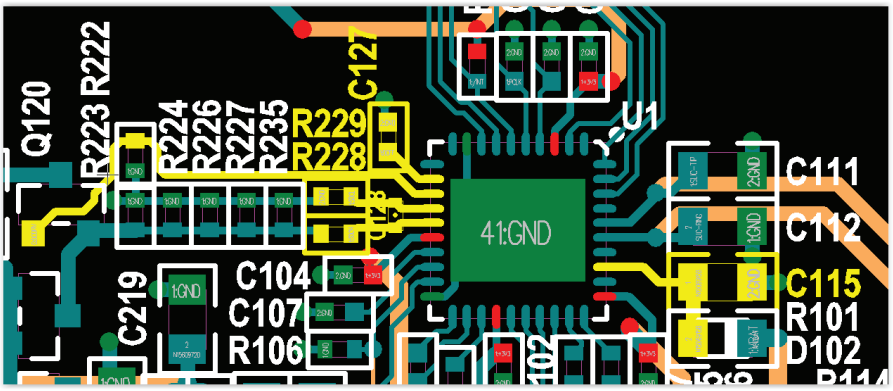


Figure 5.3. C127, C110, C115, C128 Placement

3. Minimize loop on bias reference resistor R106 from device pin to GND. R102 should be placed close to the device and can return to GND through a via if necessary.

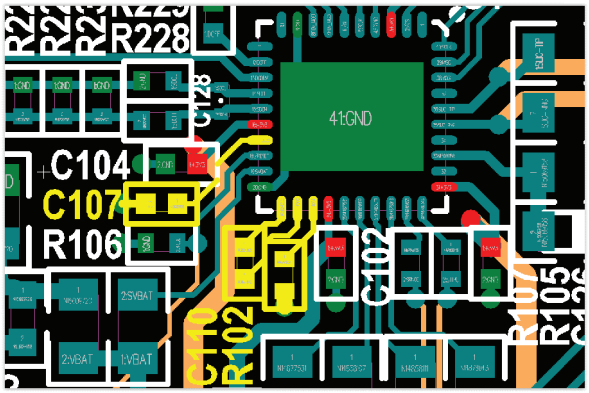


Figure 5.4. R106, R102

4. Keep first components in AC and DC sense path (C113, C114, R111, R112) as close to device as possible. Keep trace lengths of TIP and RING sense paths equal to maintain balance and improve noise immunity.

5. The coarse sense resistance is divided into two series resistors (110 kΩ and 1.47 MΩ). Keep the 110 k coarse sense resistor close to device while placing 1.47 MΩ coarse sense resistors as close to the external connection to TIP/RING as possible.

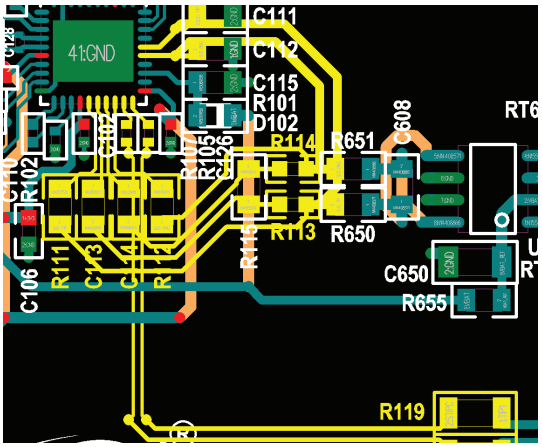


Figure 5.5. Coarse Sense Resistance

6. Route digital I/O away from analog signals. If digital trace must be routed over analog signal, route perpendicular to the analog signal to minimize coupling.

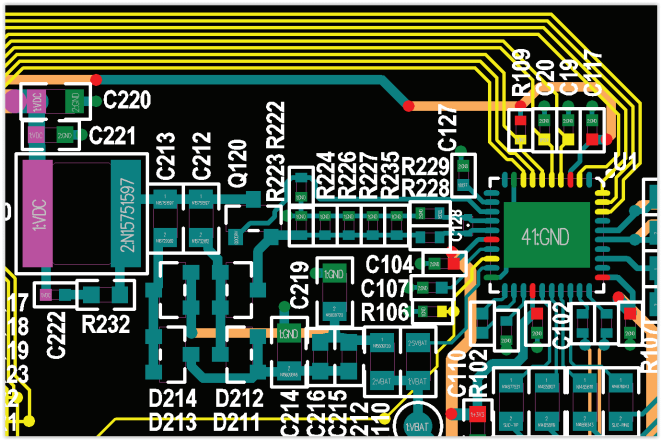


Figure 5.6. Routing Digital I/O

7. Follow the requirements for device landing pads, solder mask, solder paste, and thermal relief vias, as stated in the data sheet. Provide thermal relief metal on the bottom layer below the grounded epad of the SLIC chip. If there are more than two layers (n layers), provide more thermal relief on layer n-1 or on a ground plane.

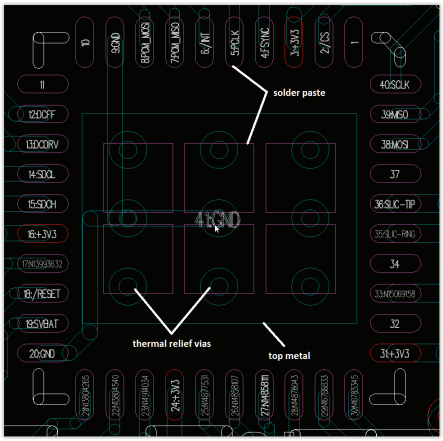


Figure 5.7. Thermal Relief Vias, Solder Paste, Top Metal

5.2 Si32182/3 Top-Level Design

This design features the ISI-enabled Si32182/3 device in a 5x6 mm 36-pin QFN package, which is shown in the figure below. Design considerations for this option include:

- Supply decoupling networks
- AC and DC sense networks
- DC-DC charge pump
- Bias network components
- Host Interface
- Optional power cross diode

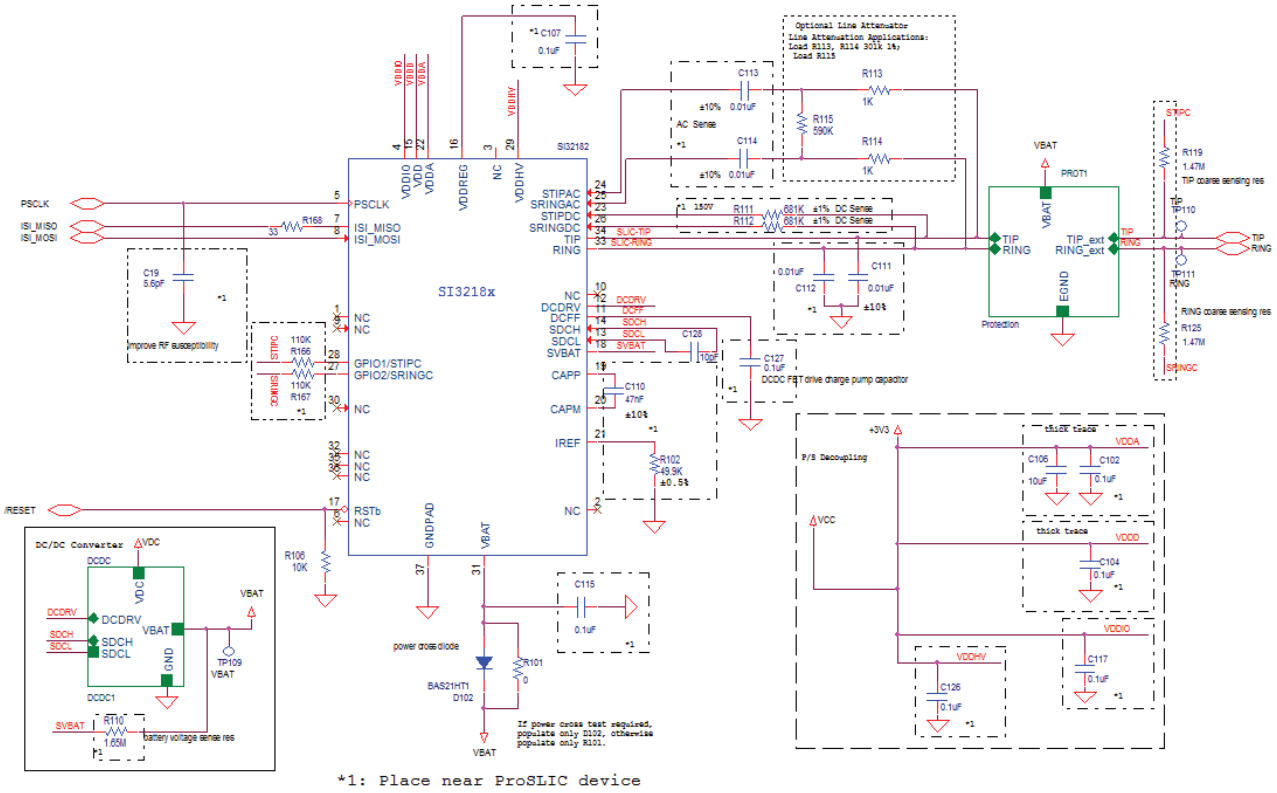


Figure 5.8. Si32182/3 Top Level Schematic

5.2.1 Si32182/3 Top-Level Bill of Materials

The table below is a comprehensive bill of materials (BOM) for the top-level Si32184/5 design.

Table 5.2. Si32182 Top-Level Bill of Materials

Quantity	Component	Description	Value	Rating	Accuracy	Type	Package
5	C102,C104,C107,C117,C126	Decoupling capacitor	0.1 µF	10 V	±10%	X7R	0402
1	C106	Decoupling capacitor	10 µF	6.3 V	±20%	X5R	0603
1	C110	Speedup capacitor	47 nF	16 V	±10%	X7R	0402
2	C111,C112	Linefeed capacitor	0.01 µF	200 V	±10%	X7R	0805
2	C113,C114	AC Sense capacitor	0.01 µF	200 V	±10%	X7R	0805
1	C115	VBAT decoupling capacitor	0.1 µF	250 V	±10%	X7T	0805
1	C127	DCDC FET drive charge pump capacitor	0.1 µF	10 V	±10%	X7R	0402
1	R101	Power cross diode bypass resistor	0	2 A		ThickFilm	0805
1	R102	Reference resistor	49.9 K	1/16 W	±0.5%	ThickFilm	0402
1	R110	VBAT sense resistor	1.65 M	1/10 W	±1%	ThickFilm	0805
2	R111,R112	DC sense resistor	681 K	1/10 W	±1%	ThickFilm	0805
2	R113,R114	AC sense resistor	1 K	1/10 W	±1%	ThickFilm	0603
2	R119,R125	Coarse sense resistor	1.47 M	1/8 W	±1%	ThickFilm	1206
2	R166,R167	Coarse sense resistor	110 K	1/16 W	±1%	ThickFilm	0402
1	R168	Signal integrity resistor	33	1/16 W	±5%	ThickFilm	0402
1	U1	SLIC IC	Si32182	100 V		MCM	QFN36 5x6mm
1	C191	Host digital I/O attenuation	5.6 pF	50 V	±0.25pF	NP0 HIGH Q	0402
1	D1022	power cross diode	BAS21HT1	200 mA		Single	SOD-323
1	R1063	Pull-down resistor	10 K	1/16 W	±5%	ThickFilm	0402
1	R1154	AC attenuator resistor	590 K	1/10 W	±1%	ThickFilm	0805
1	C1285	Current sense attenuator capacitor	10 pF	50 V	±5%	C0G	0402

Notes:

1. Optional capacitors for noise control of host interface.
2. Optional diode if design must support power cross.
3. Optional pull-down resistor for /RESET. The /RESET pin must be held low until PCM clock is established and stabilized.
4. Optional resistor for AC attenuation.
5. Optional current sense attenuator capacitor; attenuates high-frequency noise in current sense lines.

5.2.2 Si32182/3 Top-Level Layout Guidelines

Layout guidelines for the Si32184/5 also serve to guide the layout of the Si32182/3, with the addition of concern for the signals PSCLK, ISI_MOSI, and ISI_MISO, which are relatively high-speed digital signals (~24 MHz). Care should be taken to keep these signals away from sensitive analog nodes.

5.3 DC-DC Converter

The recommended DC-DC architecture for the Si3218x family is the Low-cost Capacitive Boost (LCCB). This converter is optimized for a full battery tracking architecture that meets the following power requirements.

Table 5.3. DC-DC Converter

Parameter	Description	Value
VDC	Input Voltage	4.5 V to 16 V
|VBAT|	Output Voltage Magnitude	100 V max
POUT	Average Output Power	3 W

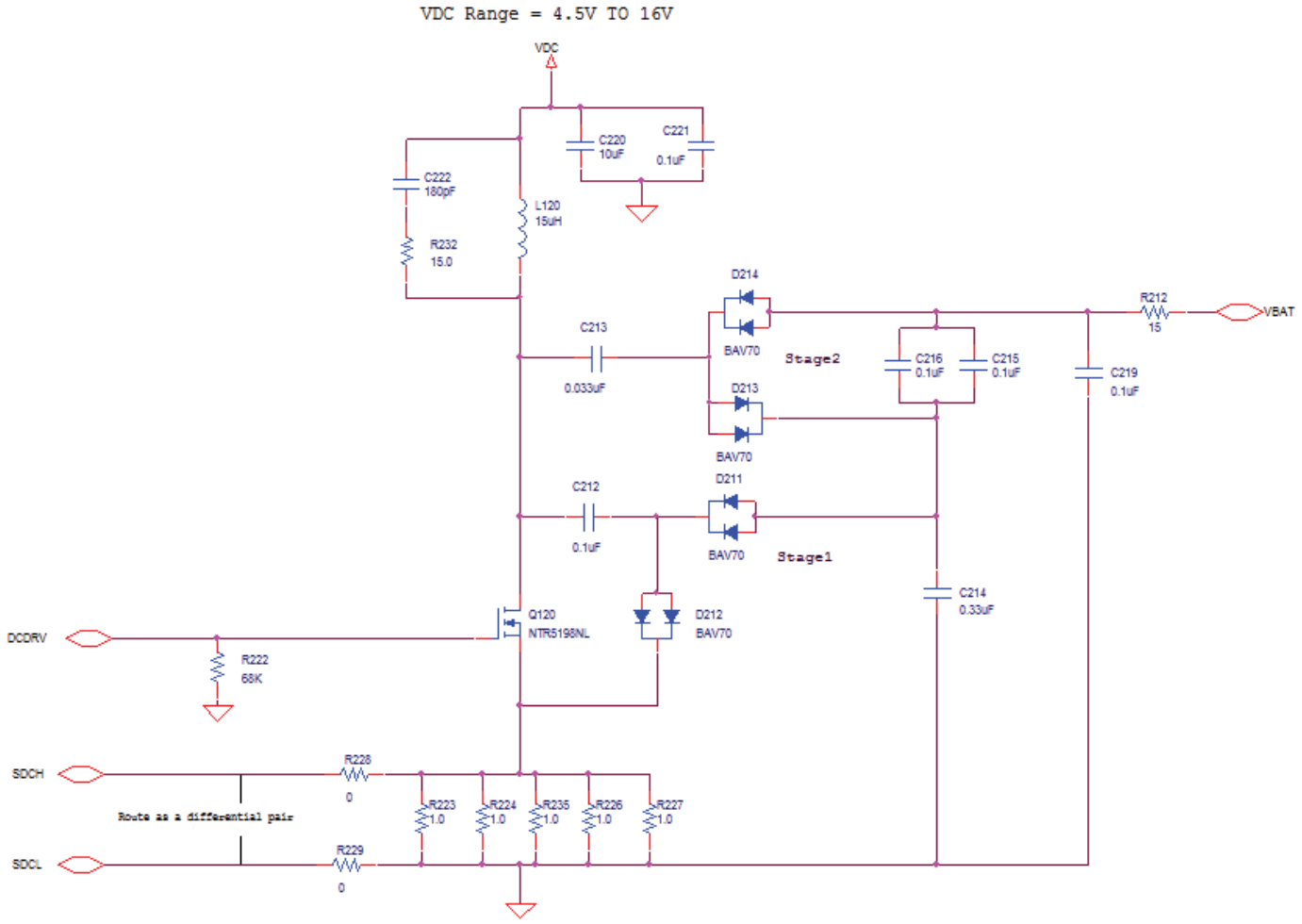


Figure 5.9. Low-Cost Capacitive-Boost DC-DC Converter

5.3.1 Bill of Materials

The table below is a comprehensive bill of materials (BOM) for the DC-DC converter. This is applicable to all designs.

Table 5.4. Low-Cost Capacitive-Boost DC-DC Converter Bill of Materials

Quantity	Component	Description	Value	Rating	Accuracy	Type	Package
1	C212	Transfer capacitor	0.1 µF	250 V	±10%	X7T	0805
1	C213	Transfer capacitor	0.033 µF	250 V	±10%	X7T	0805
1	C214	Storage capacitor	0.33 µF	50 V	±10%	X7R	0805
2	C215,C216	Storage capacitor	0.1 µF	100 V	±10%	X7R	0603
1	C219	Storage capacitor	0.1 µF	250 V	±10%	X7T	0805
1	C220	Decoupling capacitor	10 µF	16 V	±10%	X5R	0805
1	C221	Decoupling capacitor	0.1 µF	25 V	±20%	X7R	0603
1	C222	Snubber capacitor	180 pF	50 V	±5%	C0G	0402
2	D211,D214	Rectifying diode	BAV70	215 mA		Dual Common Cathode	SOT-23
2	D212,D213	Charging diode	BAV70	215 mA		Dual Common Cathode	SOT-23
1	L120	Inductor	15 µH	1.6 A	±20%	Shielded	6028
1	Q120	Current switch FET	NTR5198NL	60 V/2.2 A		N-CHNL	SOT-23
1	R212	Current limit resistor	15	1/10 W	±1%	ThickFilm	0805
1	R222	Gate resistor	68 K	1/16 W	±5%	ThickFilm	0402
5	R223,R224,R226,R227,R235[1]	Current sense resistor	1	1/16 W	±1%	ThickFilm	0402
2	R228,R229	SDCx Tap resistor	0	1 A		ThickFilm	0402
1	R232	Snubber resistor	15	1/16 W	±1%	ThickFilm	0603

Note:

1. Aggregate current sense resistor should equal 0.2 Ω (any configuration) and support 5/16 W power dissipation.

5.3.2 DC-DC Converter Layout Guidelines

1. Place the SDCH and SDCL tap resistors together with the DCDC converter current sense resistor network. Route SDCH and SDCL as a differential pair between the DCDC converter current sense resistor tap elements and the SLIC chip. Route DCDRV in parallel to SDCL, on the opposite side from SDCH. Interpose ground between these signals and other signals. Keep these signals as short as possible.

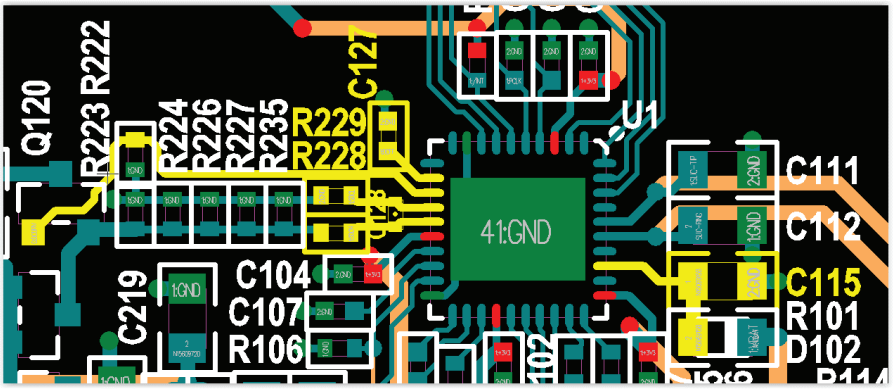


Figure 5.10. Resistor Network Layout

2. Minimize loop C220, L120, Q120, and R223-24, R226-7, R235.

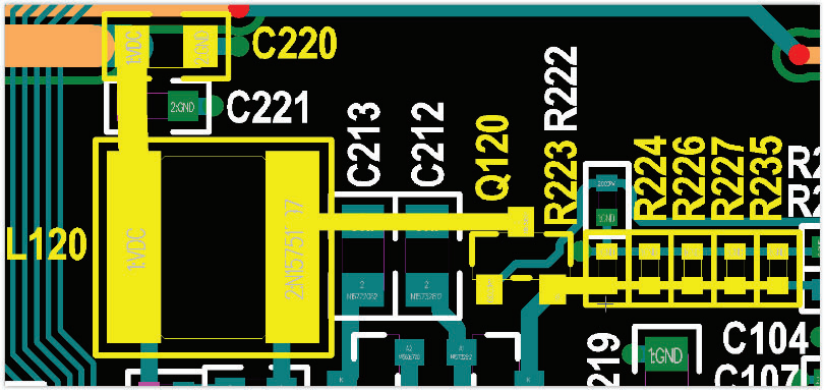


Figure 5.11. Minimizing Loop C220, L120, Q120, R223-24, R226-7, R235

3. Minimize loop C212, D212, and R223-24, R226-7, R235.

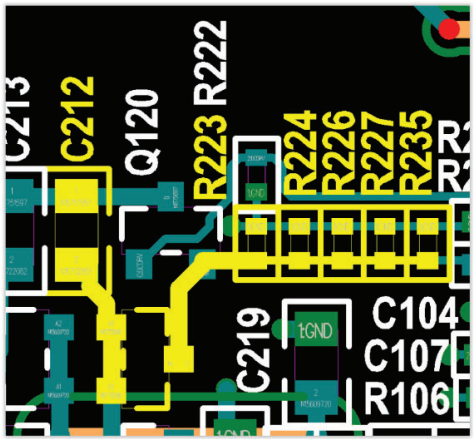


Figure 5.12. Minimizing Loop C212, D212, R223-24, R226-7, R235

4. Minimize loop C213, D214, C214, and C214-16.

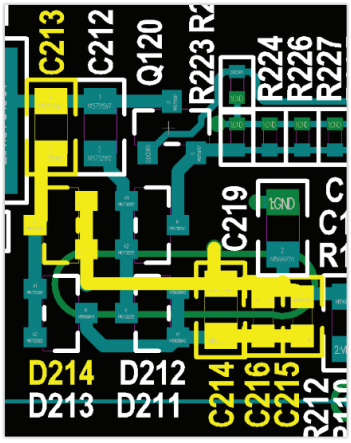


Figure 5.13. Minimizing Loop C213, D214, C214, C214-16

5. Place R211, R221, R231 close to C219.

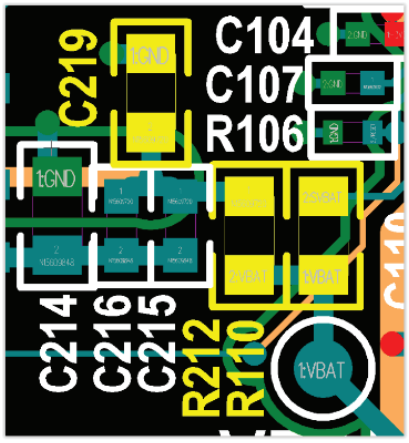


Figure 5.14. R211, R221, R231 Placement

6. Place R212 close to C215-19.

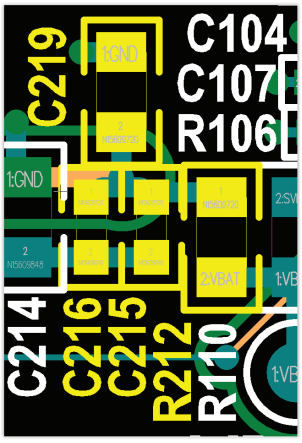


Figure 5.15. R212 Placement

5.4 Protection

Here we present a nominal protection circuit against lightning surge and over-current (power cross/contact/induction) surge. Individual application requirements will vary, and we urge you to contact your preferred protection provider for a detailed recommendation.

R650 and R651 are required. C608 is helpful for minimizing conducted emissions, but is not required. R655 is for debugging and may be replaced by a short.

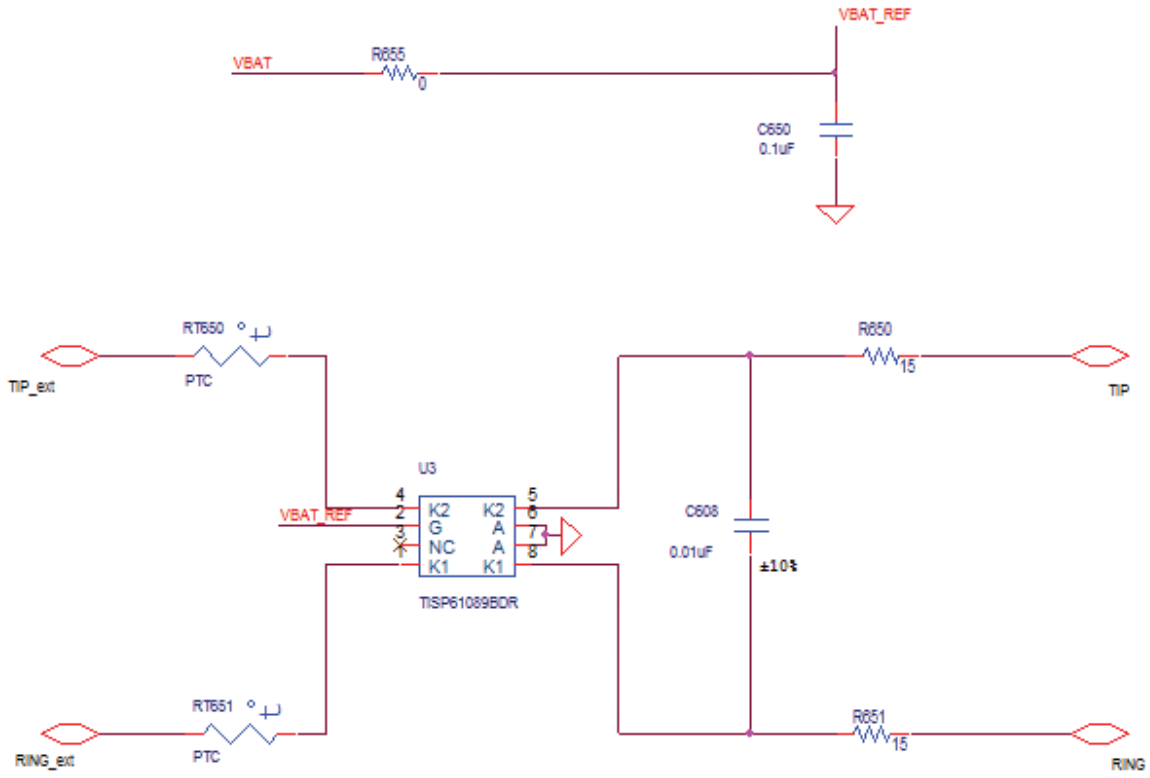


Figure 5.16. Protection Circuit Schematic

5.4.1 Bill of Materials

Table 5.5. Protection Example Bill of Materials

Quantity	Component	Description	Value	Rating	Accuracy	Type	Package
1	C608	Filter capacitor	0.01 µF	200 V	±10%	X7R	0805
1	C650	Reference decoupling capacitor	0.1 µF	200 V	±20%	X7R	1206
2	R650,R651	Current limit resistor	15	1/10 W	±1%	ThickFilm	0805
2	RT650,RT651	Overcurrent protection	PTC	3 A		TelCom	
1	U3	Overvoltage protection	TISP61089BDR	170 V		SLIC	SO8

5.4.2 Protection Layout Example

A sample protection layout is shown.

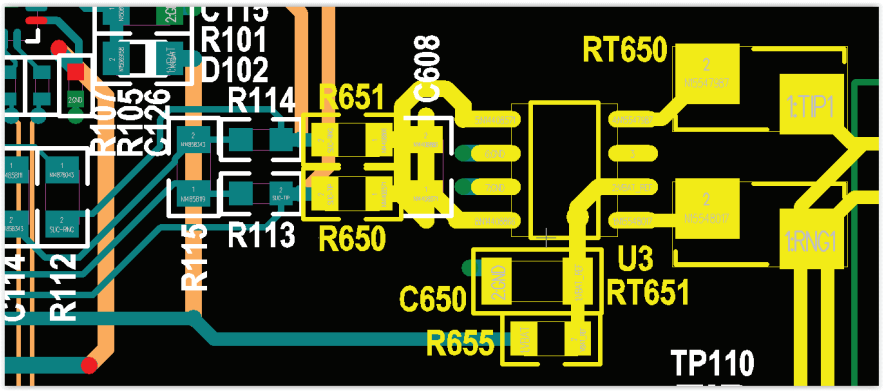


Figure 5.17. Protection

6. Design Examples

The ProSLIC API Configuration Tool is the ideal resource for translating system requirements into source code that is used by the ProSLIC API to configure the Si3218x. This section provides system-level design examples for some of the Si3218x's configurable features to demonstrate to the user how system requirements must be analyzed to ensure the proper system parameters are provided to the ProSLIC API Configuration Tool.

The system requirements that require a thorough understanding prior to using the ProSLIC API Configuration Tool are:

- Ring Generation and Ring-Trip
- DC Feed I/V

A design flow will be presented which will highlight what system parameters are required and the steps required to do the preliminary analysis prior to using the ProSLIC API Configuration Tool.

6.1 Ring Generation and Ring-Trip Design

To properly configure the Si3218x to meet the user's ringing requirements, a system-level analysis of the internal and external components of the ringing environment must be performed. This ensures the user has all information necessary to configure the Si3218x to meet their ringing requirements.

A simplified, but useful model of the Si3218x in a ringing application is shown in the figure below and the system design parameters are defined in the following table. The values of all impedances (R_{PROT}, R_{FUSE}, R_{LINE}, R_{CPE} and Z_{RINGER}) should be known prior to starting the ringing/ring-trip design process. Assume that R_S = 100 Ω.

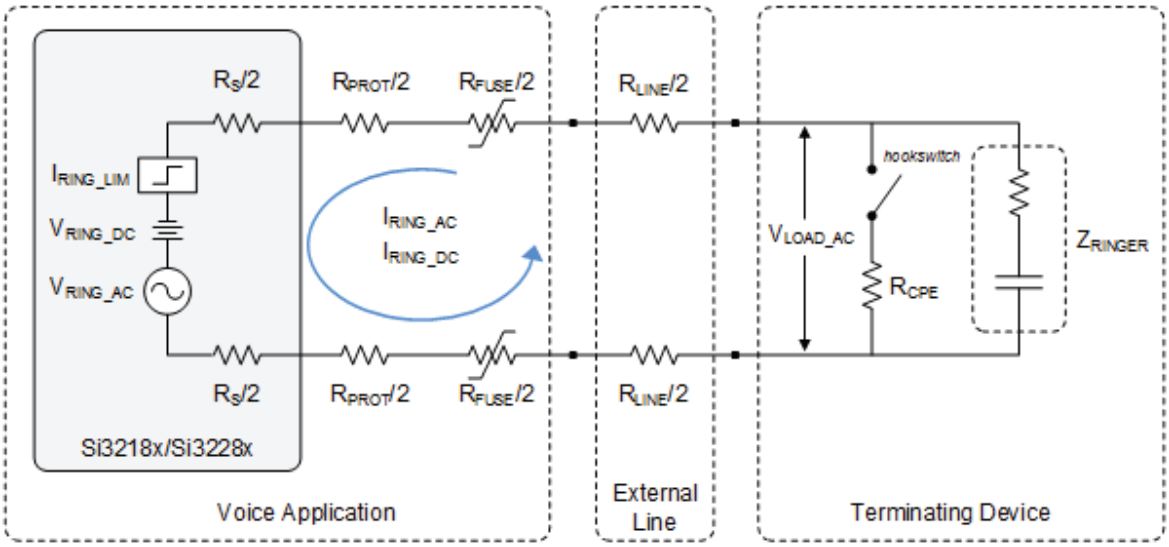


Figure 6.1. Ringing System Model, On-hook

Table 6.1. Ringing System Design Parameters

Parameter	Units	Description
V_{RING_AC}	Vrms	Programmed ringer AC voltage.
V_{RING_DC}	V	Programmed ringer DC offset.
I_{RING_LIM}	mApk	Ringer peak current limit.
V_{LOAD_AC}	Vrms	AC voltage at load when hookswitch is open and ZREN = Maximum AC load.
I_{RING_AC}	mArms	AC current flowing in loop.
I_{RING_DC}	mA	DC current flowing in loop.
R_S	Ω	Internal ringer source impedance (typically 100 Ω).
R_{PROT}	Ω	External current limiting resistor (sum of both TIP and RING resistors).
R_{FUSE}	Ω	Sum of all external fuse or PTC resistance.
R_{LINE}	Ω	Total resistance of external loop.
R_{CPE}	Ω	Resistance of off-hook CPE device (telephone).
$\lvert Z_{RINGER} \rvert$	Ω	Magnitude of AC load impedance at the ringing frequency.

No parameter has more of an impact on a voice system design than the ringing amplitude requirement. Designing to meet a maximum ringing voltage requirement will determine the maximum battery (VBAT) that is necessary, which impacts hardware component selection as well as the overall system power requirements.

Step 1: Determine Required Ringing Amplitude (V_{RING_AC}, V_{RING_DC})

Ringing amplitude requirements are typically presented by the service provider in one of two ways:

1. Open Circuit Ringing Voltage ($V_{RING_AC_OPEN}$) + Required DC Voltage (V_{RING_DC}).
2. Ringing Voltage Across the Maximum Ringer Load ($V_{RING_AC_LOAD}$) + Required DC Voltage (V_{RING_DC}).

If the required voltage is specified as the *open circuit voltage*, then:

$$V_{RING\ AC} = V_{RING\ AC\ OPEN}$$

Ringing Voltage w/ Open-Circuit Voltage is Specified

If the required voltage is specified *at the load*, the open-circuit voltage required to achieve the desired voltage at the given load must be computed:

$$V_{RING_AC} = \frac{V_{RING_AC_LOAD} \times [R_S + R_{PROT} + R_{FUSE} + R_{LINE} + |Z_{RINGER}|]}{|Z_{RINGER}|}$$

Ringing Voltage w/ Load Voltage is Specified

Now that the desired ringer source amplitude has been determined, the system must be evaluated to ensure the desired impedances (line and load) can be supported. Although this step is handled automatically by the ProSLIC API Configuration Tool, the design methodology is provided here for reference.

Step 2: Determine Max/Min On-hook and Off-hook AC Ringing Currents ($I_{PK_OFF_MAX}$, $I_{PK_ON_MIN}$)

There are two loop conditions in which the AC current during ringing must be evaluated:

Table 6.2. Critical Conditions for AC Ring-trip Evaluation

Ringing Current	Loop State	R_{LINE}	Z_{RINGER}	R_{CPE}	Total Loop Impedance (AC)
$I_{PK_ON_MAX}$	On-hook	0	minimum	N/A	Z_{ON_MIN}
$I_{PK_OFF_MIN}$	Off-hook	maximum	N/A	maximum	Z_{OFF_MAX}

When on a short loop with maximum ringer load, the on-hook AC ringing current is maximized. This value needs to be understood in order to ensure the selected DC-DC converter is capable of delivering the power to the load and to ensure that a false AC ring-trip will not occur under these conditions.

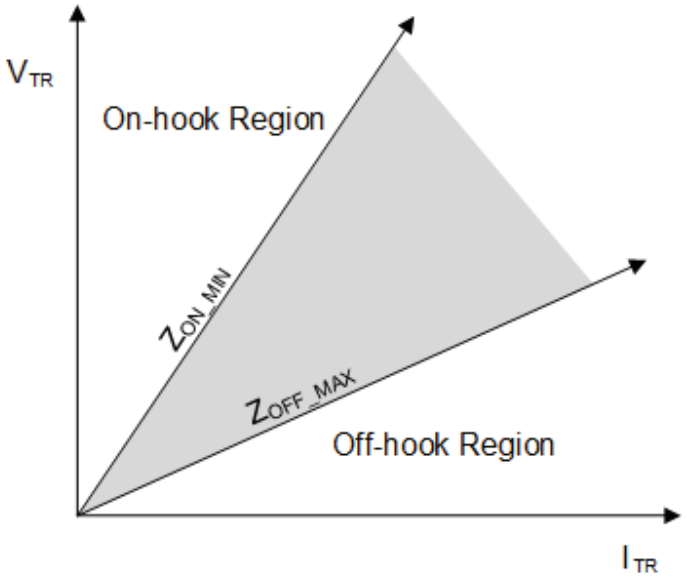


Figure 6.2. AC Ring-Trip Impedances

The minimum loop impedance and minimum ringer impedance (maximum ringer load) produce the minimum total on-hook AC loop impedance,

$$Z_{ON_MIN} = R_S + R_{PROT} + R_{FUSE} + |Z_{RINGER}|$$

Minimum On-hook AC Impedance

At this minimum impedance, the maximum AC ringing current will occur:

$$I_{PK_ON_MAX} = \frac{V_{RING_AC} \times \sqrt{2}}{Z_{ON_MIN}}$$

Maximum On-hook Ringing Current

The minimum off-hook ringing current occurs when the terminating device goes off-hook and the line and CPE resistances are at their maximums. This value is critical to understand in order to properly configure ring-trip.

$$Z_{OFF_MAX} = R_S + R_{PROT} + R_{FUSE} + R_{LINE} + R_{CPE}$$

Minimum Off-hook Ringing Current

This results in the minimum off-hook ringing current of:

$$I_{PK_OFF_MIN} = \frac{V_{RING_AC} \times \sqrt{2}}{Z_{OFF_MAX}}$$

Minimum Off-hook Ringing Current

Step 3: Determine If AC Ring-Trip Is Possible

The figure below shows the two previously calculated parameters, $I_{PK_ON_MAX}$ and $I_{PK_OFF_MIN}$, as well as two new parameters, defined as:

I_{RING_LIM}	The target ringing current clamp value. This maximum value is dependent on the DC-DC architecture and will be set between 60 and 90 mA and should not be modified by the user
I_{PK_RTACTH}	The target AC ring-trip threshold.

The target values for these parameters must be calculated as part of the AC ring-trip design.

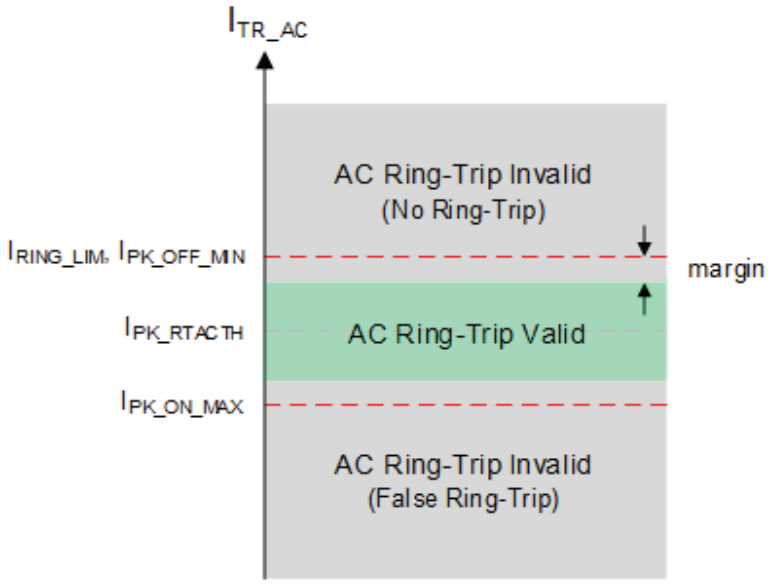


Figure 6.3. AC Ring-Trip Valid Range

To ensure that AC ring-trip may be used, the following conditions must be true:

Table 6.3. AC Ring-Trip Conditionals

Condition	Relationship	Reason
AC1	$I_{PK_OFF_MIN} > I_{PK_ON_MAX}$ + 20%	Ensures AC ring-trip can be achieved with adequate margin.

Condition	Relationship	Reason
AC2	$I_{PK_ON_MAX}$ + 20% < I_{RING_LIM}	Ensures adequate current clamp margin.

Note: It is possible (and very common) for $I_{PK_OFF_MIN}$ to be greater than I_{RING_LIM}. In this case, the loop current during ringtrip will be limited by I_{RING_LIM}.

If AC ring-trip can be used, then a threshold value should be selected between the maximum on-hook ringing current and the ringing current clamp.

$$I_{PK_RTACTH} = \frac{I_{PK_ON_MAX} + I_{RING_LIM}}{2}$$

AC Ring-Trip Threshold

Step 4: Determine if DC Ring-Trip is Possible

If conditions for AC ring-trip cannot be met or the user prefers to add DC offset to the ring signal and use DC ring-trip, the following procedure can be used to determine if DC ring-trip can be used.

During ring-trip, if the ringing signal contains a DC offset, there will be a resulting DC current flowing in the loop. The worst case (minimum) DC current that will flow will be with the maximum CPE off-hook impedance on the maximum (longest) loop. The appropriate offset voltage, VRING_DC, should be chosen such that the DC current will be:

$$I_{DC_OFF_MIN} = \frac{V_{RING_DC}}{Z_{OFF_MAX}} \geq 10mA$$

Minimum DC Ring-Trip Current

While the CPE is on-hook, no DC current should flow during ringing. However, due to offset currents, ringing distortion and transient events, margin should be added to prevent false DC ring-trip events while on-hook. To accommodate this, the minimum DC ring-trip current should be:

$$I_{DC_RT_MIN} = 5mA$$

Maximum On-hook DC Current During Ringing

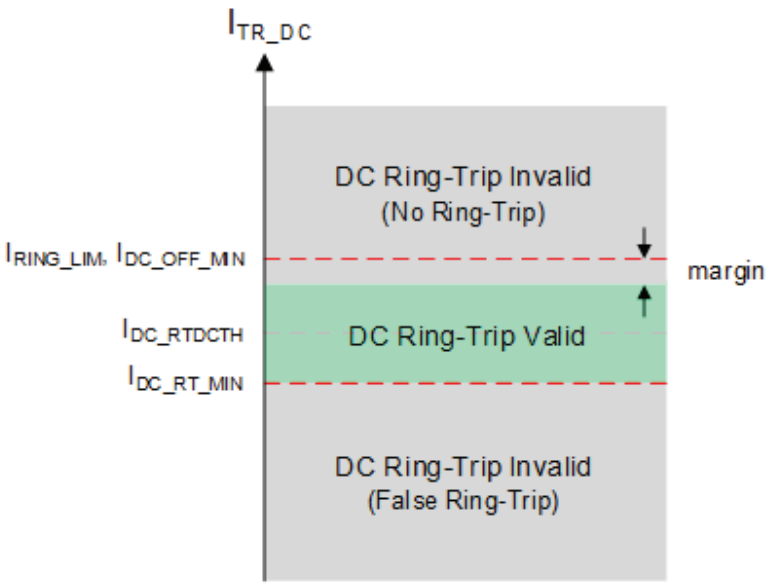


Figure 6.4. DC Ring-Trip Valid Range

To ensure DC ring-trip may be used, the following conditions must be true:

Table 6.4. DC Ring-Trip Conditions

Condition	Relationship	Reason
DC1	$I_{DC_OFF_MIN} > I_{DC_RT_MIN}$ + 3 mA	Ensures DC ring-trip can be achieved with adequate margin above minimum allowed setting.
DC2	$I_{DC_OFF_MIN}$ + 10 mA < I_{RING_LIM}	Ensures adequate current clamp margin.

If DC ring-trip can be used, then a threshold value should be selected between the maximum on-hook ringing current and the ringing current clamp:

$$I_{DC_RTDCTH} = \frac{I_{DC_OFF_MIN} + I_{DC_RT_MIN}}{2}$$

DC Ring-Trip Threshold

Step 5: Determine Ring-Trip Debounce Timers

For AC or DC ring-trip, the debounce timers should be programmed to an integer number of ½-periods of the ringing waveform. It is recommended that three times the ½-period of the ringing signal be used.

For example, for a 20Hz ringing signal, the recommended debounce time will be:

$$t_{DEBOUNCE} = 3 \text{ x } \frac{1}{2} \text{ x } \frac{1}{20 \text{ Hz}} = 75 \text{ ms}$$

6.2 DC Feed Design

To properly configure the DC feed on the Si3218x, the parameters listed in the table below will need to be determined.

Table 6.5. DC Feed System Parameters

Parameter	Units	Description	System Requirement	Hardware Dependency
V_{OC}	V	Required open-circuit voltage when on-hook.	X	
I_{SC}	mA	Required loop current when off-hook.	X	
R_{FUSE}	Ω	Series resistors on TIP and RING.		X
R_{PROT}	Ω	PTC or Fuse resistance.		X
R_{LOOP}	Ω	DC impedance loop at maximum length.	X	
R_{CPE}	Ω	Minimum DC impedance of terminating CPE device.	X	

Referring again to the basic DC feed I/V model in Figure 2.16 Si3218x DC Feed I/V Characteristics on page 15, we can generate all critical parameters on the DC feed I/V based on the system requirements and hardware dependencies provided in the table above. These parameters are then used by the ProSLIC API Configuration Tool to generate DC Feed configuration presets.

1. **Calculate the Maximum Loop Voltage Supported at ISC**

The maximum theoretical loop voltage that can be supported at the target loop current is dependent on the minimum slope requirement of the DC feed, which is 160 Ω. Assuming both the Constant-V and Resistive regions of the DC feed are set to this minimum (see Figure 6.5 DC Feed I/V Maximum Loop on page 166), the maximum possible V_{ILIM} can be calculated:

$$V_{ILIM_MAX} = V_{OC} - (I_{SC} \times 160)$$

While on the maximum loop length supported, the total off-hook impedance will be:

$$R_{OFF} = R_{PROT} + R_{FUSE} + R_{LOOP} + R_{CPE}$$

If the system must deliver the specified loop current ISC with the maximum off-hook impedance, R_{OFF}, then V_{ILIM} must be large enough such that the DC feed remains in the Constant-I region. To calculate the required minimum value of V_{ILIM} to deliver I_{SC} to the maximum off-hook impedance $R_{OFF_}$.

$$V_{ILIM_REQ} = I_{SC} \times R_{OFF}$$

If V_{ILIM_REQ} is less than V_{ILIM_MAX}, then the desired I_{SC} can be delivered on the maximum loop impedance and V_{ILIM} should be set as:

$$V_{ILIM} = \frac{V_{ILIM_{MAX}} + V_{ILIM_{REQ}}}{2}$$

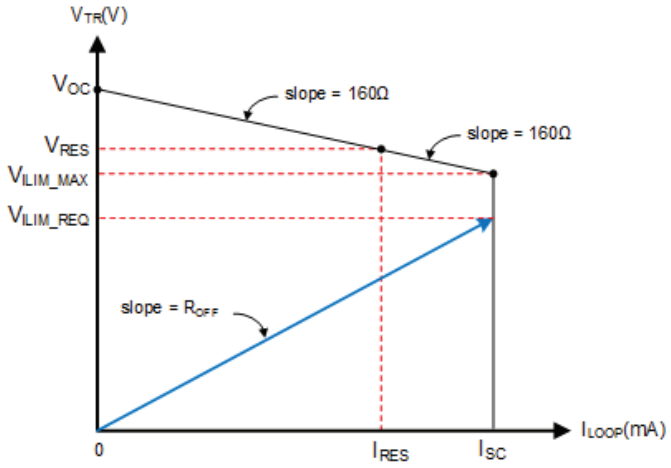


Figure 6.5. DC Feed I/V Maximum Loop

2. **Calculate V_{RES} and I_{RES}**

As long as the slopes of the Constant-V and Resistive regions are greater than 160 Ω, the selection of V_{RES} and I_{RES} is somewhat arbitrary. One method is to select a value for I_{RES} that is 80% of I_{SC}, presume the Constant-V region to be 160 Ω and calculate V_{RES} from there.

$$I_{RES} = 0.8 \times I_{SC}$$

V_{RES} may be calculated as:

$$V_{RES} = V_{OC} - (I_{RES} \times 160)$$

Now, all critical DC feed I/V parameters are known and the user may enter these into the ProSLIC API Configuration Tool as summarized in the following table.

Table 6.6. DC Feed Configuration Tool Design Inputs

ProSLIC API Configuration Tool DC Feed Input	Design Parameter
V_VLIM	V_{OC}

ProSLIC API Configuration Tool DC Feed Input	Design Parameter
V_RFEED	V_{RES}
V_ILIM	V_{ILIM}
I_VILIM	0
I_RFEED	I_{RES}
I_ILIM	I_{SC}